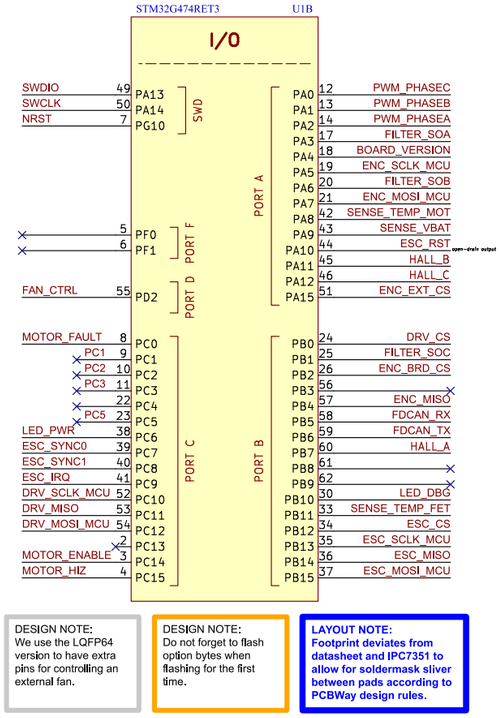
[5] MCU - I/Os

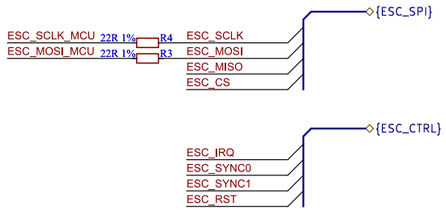


DESIGN NOTE:
We use the LQFP64 version to have extra pins for controlling an external fan.

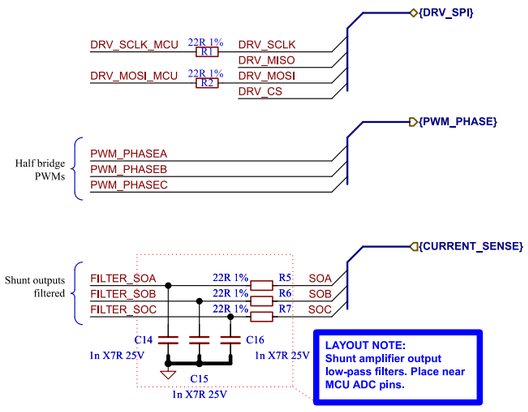
DESIGN NOTE:
Do not forget to flash option bytes when flashing for the first time.

LAYOUT NOTE:
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

Microcontroller I/O Domain

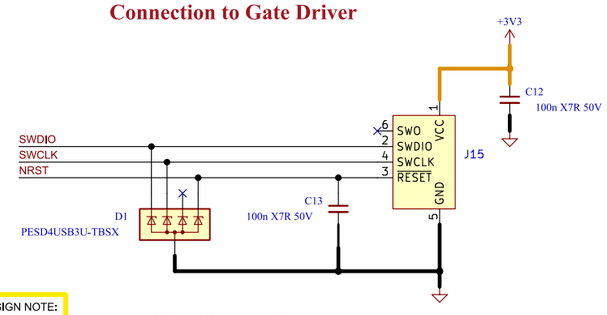


Connection to EtherCat Subdevice Controller



LAYOUT NOTE:
Shunt amplifier output low-pass filters. Place near MCU ADC pins.

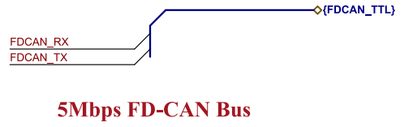
Connection to Gate Driver



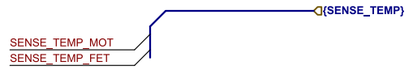
DESIGN NOTE:
Ideally TVS should be connected to chassis but makes it harder to route.

DESIGN NOTE:
Only connect GND if connected device does not share ground!

SWD (Debug) Port



5Mbps FD-CAN Bus



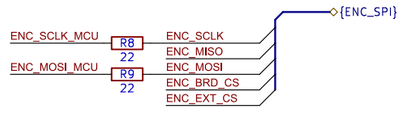
Temperature Sense

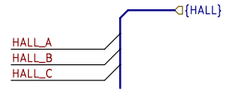


Battery Voltage Sense



Low-Side Fan Control

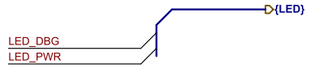




Position Sense



Board Version Selection



User LEDs

Gate Driver

MOTOR_ENABLE	TP1	MOTOR_ENABLE
MOTOR_HIZ	TP2	MOTOR_HIZ
MOTOR_FAULT	TP3	MOTOR_FAULT
DRV_SCLK	TP4	DRV_SCLK
DRV_MISO	TP5	DRV_MISO
DRV_MOSI	TP6	DRV_MOSI
DRV_CS	TP7	DRV_CS
PWM_PHASEA	TP8	PWM_PHASEA
PWM_PHASEB	TP9	PWM_PHASEB
PWM_PHASEC	TP10	PWM_PHASEC
SOA	TP11	SOA
SOB	TP12	SOB
SOC	TP13	SOC

Debug

SWDIO	TP14	SWDIO
SWCLK	TP15	SWCLK
NRST	TP16	NRST
ESC_RST	TP17	ESC_RST
ESC_SCLK	TP18	ESC_SCLK
ESC_MISO	TP19	ESC_MISO
ESC_MOSI	TP20	ESC_MOSI
ESC_CS	TP21	ESC_CS
ESC_IRQ	TP22	ESC_IRQ
ESC_SYNC0	TP23	ESC_SYNC0
ESC_SYNC1	TP75	ESC_SYNC1

RS-422

ENC_SCLK	TP24	ENC_SCLK
ENC_MISO	TP25	ENC_MISO
ENC_MOSI	TP26	ENC_MOSI
ENC_BRD_CS	TP27	ENC_BRD_CS
ENC_EXT_CS	TP28	ENC_EXT_CS

LEDs

LED_DBG	TP31	LED_DBG
LED_PWR	TP32	LED_PWR

FD-CAN

FDCAN_RX	TP33	FD_CAN_RX
FDCAN_TX	TP34	FDCAN_TX

Fan

FAN_CTRL	TP35	FAN_CTRL

Sense

SENSE_TEMP_MOT	TP36	SENSE_TEMP_MOT
SENSE_TEMP_FET	TP37	SENSE_TEMP_FET
SENSE_VBAT	TP38	SENSE_VBAT

Test Points

Comments: **References:** Flexible I/O worked examples Flexible I/O source configuration	Company: **ETH Robotics Club**		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: MCU - I/Os	File Name: MCU - IOs.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - IOs/		Reviewer:	Size: **A3**	Sheet: **5** of **20**